

Beyond Dominant-Strategy: Bayesian Settings

There are many reasons why we can't always require dominant strategies when design mechanisms.

- (1) Requiring such a strong concept might not be tractable.
- (2) Agents do not always have dominant strategies! What then?

Use *dominant strategy* whenever you can get it—it needs no assumption about what bidders believe, and Myerson's Lemma hands you the payments. Reach for *Bayes-Nash* only when the mechanism you care about has no dominant strategy (first-price, all-pay), or when DSIC provably can't achieve what you want. The price you pay is a modeling assumption: everyone knows $\mathbf{F}$, and everyone plays the equilibrium.

We'll now introduce the Bayesian setting.

Suppose the valuation v_i of bidder i is drawn from a prior distribution F_i . What do the following symbols mean? Give a formula for each of the first two.

- $F_i(x) =$
- $f_i(x) =$
- $\mathbf{F}$ or $\vec{F}$:

Unless otherwise noted, we assume that the prior distribution $\mathbf{F}$ is *common knowledge* to all bidders and the mechanism designer (the seller).

Definition 1. A *Bayes-Nash equilibrium (BNE)* for a joint distribution $\mathbf{F}$ is a strategy profile $\sigma = (\sigma_1, \dots, \sigma_n)$ such that for all i and v , $\sigma_i(v_i)$ is a best-response when other agents play $\sigma_{-i}(\mathbf{v}_{-i})$ when $\mathbf{v}_{-i} \sim \mathbf{F}_{-i} \mid v_i$.

First-Price Auction: Bayes-Nash Equilibrium

Claim 1. Consider two identically and independently drawn bidders from $F = U[0, 1]$. It is a (symmetric) BNE for each bidder to bid $\sigma_i(v_i) = v_i/2$ in the first-price auction.

Fixed, choice, and randomness. Before proving the claim, identify the following for bidder 1's best response:

- a. What is fixed?
- b. What does bidder 1 choose?
- c. What is random in bidder 1's expected utility?

Proof.

Bayesian Settings

Using notions from the Bayesian setting and how bidders Bayesian update as they learn information, we define three stages of the auction:

1. *ex ante*:

2. *interim*:

3. *ex post*:

Typically we discuss the *ex post* allocation and payment rules as a function of all of the values. However, in the Bayesian setting, to reason about BIC, it often makes sense to take in terms of *interim* allocation and payment rules which have the same information as bidder i before the auction is run.

Definition 2. We define the *interim* allocation and payment rules in expectation over the updated Bayesian prior given i 's valuation:

$$x_i(v_i) =$$

and

$$p_i(v_i) =$$

Pitfall: we are now overloading x_i . The *ex post* $x_i(\mathbf{v})$ takes the whole profile and returns 0 or 1; the *interim* $x_i(v_i)$ takes one number and returns a probability in $[0, 1]$. Same letter, different function—when you read a formula, first ask which one you're looking at.

Theorem 1 (Revenue Equivalence). *The payment rule and revenue of a mechanism is uniquely determined by its allocation. Hence, any two mechanisms with the same allocation must earn the same expected revenue in Bayes-Nash Equilibrium.*

What is this theorem a corollary of? Prove this for the first-price auction and the Vickrey (second-price) auction in the above setting!

Proof.

Our definition of Bayesian Incentive-Compatibility then follows:

Definition 3. A mechanism with *interim* allocation rule x and *interim* payment rule p is Bayesian Incentive-Compatible (BIC) if

$$v_i x_i(v_i) - p_i(v_i) \geq v_i x_i(z) - p_i(z) \quad \forall i, v_i, z.$$

Virtual Welfare

Imagine a single buyer will arrive with their private value v . We want to design DSIC mechanisms.

What mechanism should you use to maximize *welfare*?

What should you do to maximize (expected) revenue?

Definition 4. In a deterministic mechanism, given other bids $\mathbf{b}_{-i}$, bidder i 's *critical bid* is the minimum bid $b_i^* = \min\{b_i : x_i(b_i, \mathbf{b}_{-i}) = 1\}$ such that bidder i is allocated to.

Then with $\mathbf{b}_{-i}$ fixed, for all winning $v_i \geq b_i^*$, i 's payment $p_i(v_i, \mathbf{b}_{-i}) = b_i^*$ is their critical bid.

What is winner i 's critical bid in a single-item auction?

What about in the k identical item setting?